

Git Pull vs Git Fetch

Introduction

Both `git pull` and `git fetch` are used to update your local repository with changes from a remote repository, but they work in different ways.

 **Watch the Video:**

<https://youtu.be/xvU1zboOvj0?si=dsT7kWFjThk0LKGv>

What is Git Fetch?

`git fetch` downloads the latest changes from the remote repository but **does not merge them** into your current branch.

Command:

```
git fetch origin
```

What it does:

- Downloads new commits
- Updates remote tracking branches
- Does NOT change your working code

What is Git Pull?

`git pull` is a combination of **fetch + merge**. It downloads changes and automatically merges them into your current branch.

Command:

```
git pull origin main
```

What it does:

- Downloads changes from remote
- Automatically merges into local branch
- Updates your working directory

Difference Between Git Pull and Git Fetch

Git Fetch	Git Pull
Only downloads changes	Downloads + merges changes
Does not modify working code	Updates working code immediately
Safe to review changes first	Directly applies changes
Manual merge required	Auto merge

When to Use

Use Git Fetch when:

- You want to review changes before applying
- Working on important or large projects
- Avoiding automatic merge conflicts

Use Git Pull when:

- You want quick updates
- Working on small or personal projects
- You trust remote changes

Key Points

- `git fetch` = download only
- `git pull` = download + merge
- Fetch is safer for reviewing changes
- Pull is faster but less controlled
- Both help keep your repository updated